Prashant Bhandari

PCB Design Engineer | Embedded Firmware Developer
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LinkedIn | **GitHub** | **Portfolio Website**

Summary

Electronics Engineering student specializing in PCB design, embedded firmware development, and robotics systems. Experienced in designing hardware and firmware for embedded products using ESP32, Arduino, MicroPython, and C/C++.

Technical Skills

PCB Design: KiCad, Proteus, Schematic Design, PCB Layout
Embedded Systems: ESP32, Arduino, ECU2000, HCM111Z, BLE Communication
Programming: C/C++, Python, MicroPython, JavaScript
Hardware Development: Buck/Boost, IoT, QC, Robotics Systems, BLE devices

Education

B.E. Electronics, Communication & Information Engineering
Pashchimanchal Campus, Pokhara (2023 – Present, 4th Year)

Professional Experience

Electronics Design Intern – SEDS Nepal (Remote) | Sept 2025 – Jan 2026
- Designed electronics modules for pseudo satellite systems
- Performed component selection and budgeting
- Ensured modular and reproducible system architecture

Electronics Engineer Intern – YarsaTech, Pokhara | Jan 2025 – July 2025

- Designed and optimized buck converter circuits
- Developed automated QC testing device for power backup devices
- Built BLE-enabled QR/barcode scanner systems
- Implemented firmware using MicroPython and C/C++

Leadership

Electronics & Software Lead – Robotics Club, Pashchimanchal Campus (2024 – 2026)
- Mentored 350+ students in embedded systems and PCB design
- Initiated In-house PCB prototyping lab
- Led robotics competition projects
- Organized technical workshops and training programs

Projects

Solar Based Real Time Greenhouse Automation System
Developed a solar-powered real-time greenhouse automation system using an ESP32 microcontroller to monitor soil moisture, temperature and humidity. Implemented automated irrigation and mist control using sensors, MOSFETs with BLE-based data transmission and an OLED dashboard for real-time monitoring, improving water efficiency and reducing manual greenhouse management.

Pocket Cube Satalite Module
Designed and built a 1.5U PocketQube educational CubeSat prototype, integrating mechanical structure, stacked PCB architecture, and core subsystems including OBC, power management, sensors, and LoRa communication. Used a NodeMCU microcontroller interfaced with environmental and motion sensors, powered by a Li-Po battery with solar charging, demonstrating compact satellite system integration.

LoRaNet Communication Module

Designed a long-range wireless communication system using LoRa technology for environments without cellular or internet connectivity. Developed schematic and PCB using KiCad, implemented embedded firmware for packet transmission and integrated LoRa transceivers with microcontroller (Esp32) interfaces for reliable long-distance communication gained approximate 7km range while making the line of sight.

Smart Helmet Safety System

Developed an IoT-based smart helmet prototype for health monitoring and accident detection. Features include pulse rate monitoring, SpO2 measurement, body and ambient temperature sensing, fall detection, and SOS emergency alert system. Data is transmitted via BLE to a monitoring dashboard to the supervisor.

DC to DC Buck Convertors

Designed and implemented a 12V to 5V DC Buck Converter using the XL2013E1 controller, including schematic design, PCB layout, and iterative hardware optimization in KiCAD. Performed component calculations, ripple and efficiency analysis, and improved layout across multiple iterations to achieve ~ 90% efficiency, reduced ripple voltage, and stable regulated output suitable for embedded and industrial applications.

Micromouse Autonomous Maze Solving Robot

Designed and built an autonomous robot capable of solving complex mazes using PID-based motion control and sensor feedback. Created custom PCB reducing board size by 40% and achieved 95% navigation accuracy using QTR sensor array and optimized algorithms.

Voyager - Line Follower Robot

Developed a competition-grade line follower robot optimized for high speed and stability. Implemented PID-based line tracking and ultrasonic obstacle detection for improved navigation.

Autonomous Boat Navigation System

Designed electronics and control architecture for an autonomous racing boat including propulsion control, sensor integration, and embedded control firmware.

Achievements

Startup Innovation Recognition | Safety Modules | 2026

Selected among the Top 25 teams nationwide in the U.S.–Nepal Startup Weekend Challenge.

International Representation | Safety Modules | 2025

Selected as a finalist representing Nepal at Techfest IIT Bombay.

Micromouse Competitions | Line maze Solving Robot | 2024 – 2025

1st Place – LOCUS 2025, Pulchowk Campus, Kathmandu Nepal (Institute of Engineering)
2nd Place – Delta 5.0, Purwanchal Campus, Dharan Nepal (Institute of Engineering)
5th Place – Techfest **IIT Bombay**, Mumbai, India

Autonomous Boat Racing Competition | Robotics Club WRC | 2024

1st Runner-Up – Designed navigation electronics and control system.

Line Follower Robotics Competition | Robotics Club WRC | 2023

1st Place – Developed high-speed line follower robot.

Volunteering

Robotics Mentor – Karyashala & Engineers Without Borders (2024)

Trained 50+ students in ESP32 systems, 3D Design, 3D Printing, and Robotics development.